

62
Sm

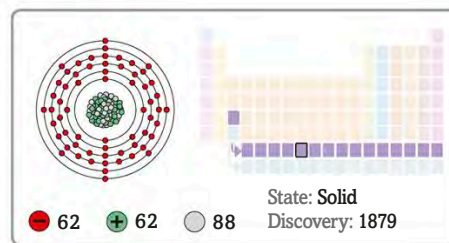
Samarium

This element is named after the mineral **samaraskite** from which it was first purified. However, another lanthanide-rich mineral called monazite is the main source of this element today. Samarium is mixed with cobalt to make permanent magnets that are often used in **electric guitars**.

This silvery white metal darkens on contact with air.

These pickups (components that sense vibrations produced by guitar strings) are made of samarium-cobalt magnets.

Laboratory sample of pure samarium



Guitar pickups

Lanthanides

63
Eu

Europium

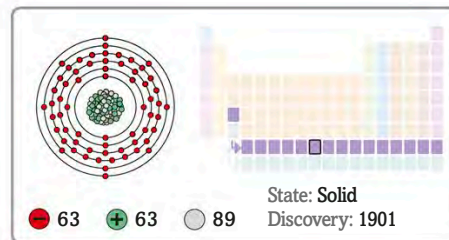
Laboratory sample of pure europium



Europium was named after the continent of **Europe**. However, most of the world's supply of the element comes from the USA and China, where the mineral bastnasite is mined for the extraction of **pure europium**. A compound called europium oxide is used in euro and **British bank notes**. When placed under ultraviolet (UV) light, the compound gives off a red glow.

This red glow proves this note is real.

The crystals of this yellowish metal often have patches of dark oxides.



Section of British note under UV light